

EDUCATION

Beijing Normal University 985 211 Double 1st-Class

Aug 2023 - Jul 2026

Teaching Chinese to Speakers of Other Languages Master

Hefei University of Technology 211 Double 1st-Class

Sep 2018 - Jul 2022

New Energy Materials and Devices Bachelor

PUBLICATION

Transcranial photobiomodulation on the left inferior frontal gyrus enhances speech production efficiency

Guang Yang, Zhaoqian Yue, Mingchuan Yang, Dongwei Li, Luyao Chen

- Refereed Journal Article: *Scientific Reports*
- 1st Author
- Not yet published -Under review

Transcranial Ultrasound Stimulation Unravels the Causal Role of the Left Inferior Frontal Gyrus in Syntactic Processing

Luyao Chen, Guang Yang, Xingfang Qu, Dongwei Li, Chenguang Zhao, Junjie Wu, Ke Zeng, Robert Chen, Angela D. Friederici

- Refereed Journal Article: *Nature Communications*
- 1st Author (co-first author, supervisor as first author)
- Not yet published -Writing in progress

Transcranial photobiomodulation on the left inferior frontal gyrus enhances Mandarin Chinese L1 and L2 complex sentence processing performances

Mingchuan Yang, Yang Liu, Zhaoqian Yue, Guang Yang, Xu Jiang, Yimin Cai, Yuqi Zhang, Xiujie Yang, Dongwei Li, Luyao Chen

- Refereed Journal Article: *Brain and Language*
- Other Writing – review & editing
- Published

Theta oscillations underpin syntactic encoding in first and second language sentence production

Zhaoqian Yue, Shuning Tang, Mingchuan Yang, Guang Yang, Peisong Yan, Dongwei Li, Angela D. Friederici, Luyao Chen

- Refereed Journal Article: *PNAS*
- Other Writing – review & editing
- Not yet published – Under review

RESEARCH PROJECTS

National Social Science Fund Youth Project: Neural Mechanisms of Chinese Grammar Processing and Learning Based on Neural Localization and Modulation Technologies

Sep 2023 - Dec 2025

National Project

- Conducted a tPBM experiment on second-language learners, demonstrating causal enhancement of syntactic learning efficiency through LIFG modulation.
- Led a TUS experiment on native Chinese speakers, revealing inhibitory effects of LIFG stimulation on hierarchical syntactic computation.

National 2030 Major Science and Technology Project: Brain Science and Brain-inspired Intelligence: Chinese School-age Children Brain and Cognitive Development Cohort Study

Sep 2023 - Present

National Project

- Processed large-scale fMRI data, performed motion correction, normalization, and smoothing, and conducted functional connectivity analyses using CONN and graph-theoretical metrics to examine inter-regional communication efficiency; implemented automated pipelines for data quality assurance and cross-site standardization.

Development of a Multimodal Data: based Assessment Tool for Contextual Communication Ability National Project Participated in material design, program implementation, data analysis, and manuscript drafting	Sep 2023 - Oct 2024
Max Planck Partner Group Project: Universal Neural Substrates for Category-based Hierarchical Syntactic Processing in Language International Cooperative Project Participated in constructing syntactic materials controlled for lexical and semantic confounds, programmed the bilingual experiment (Chinese–English) using E-prime, and conducted RSA and MVPA analyses to explore representational patterns across languages. Collaborated remotely with the Leipzig MPI-CBS team on model interpretation and figure preparation for an upcoming journal submission.	Sep 2023 - Dec 2025
Near-infrared Laser Stimulation for Improving Developmental Dyslexia in Children: Mechanisms and Effects Provincial Project Designed and optimized near-infrared stimulation protocols for children, conducted pre- and post-intervention behavioral assessments, managed participant scheduling and data recording, and analyzed cognitive performance improvements to evaluate the effectiveness of laser-based intervention in developmental dyslexia.	Nov 2023 - Present
BNU Education Foundation Project: Neural Mechanisms of Cross-Domain Multi-Level Analogical Reasoning University-Level Project Involved in material and program development, data analysis, and manuscript preparation.	Sep 2023 - Dec 2024
BNU Young Faculty Fund: Neural Mechanisms of Functional-word-based Chinese Syntactic Merging and Learning Processes University-Level Project Developed and refined sentence materials emphasizing functional-word cues, designed behavioral tasks to examine syntactic merging and learning, conducted data collection and statistical analysis, and summarized results for internal reports and manuscript preparation.	Sep 2023 - Dec 2024

MISCELLANEOUS

- Programming and Data Analysis Skills:** MATLAB (SPM, DPABI, CONN), Python (RSA, MVPA), R (LME, DDM), E-Prime, PsychoPy, Linux shell scripting.
- Languages:** Mandarin Chinese (Putonghua Proficiency Test, Grade II–A); English (IELTS 6.5); Italian (Graduate-level elective course)